



CHEMISTRY CH.5 — METALLURGY — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Important Ores of Metals
2. Concentration & Dressing of Ores
3. Roasting, Calcination & Smelting
4. Iron & Steel Production
5. Aluminium Extraction
6. Copper Extraction
7. ⚠ Traps / Key Differences Summary (Q519–Q531)

1. Important Ores of Metals

- Aluminium ke ores: Bauxite ($\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$), Corundum (Al_2O_3), Cryolite (Na_3AlF_6). [Q519]
- Iron ke ores: Hematite (Fe_2O_3), Magnetite (Fe_3O_4), Siderite (FeCO_3), Limonite ($2\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$). [Q519, Q521]
- Copper ke ores: Copper pyrites (CuFeS_2), Copper glance (Cu_2S), Malachite [$\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$]. [Q521]
- Zinc ke ores: Zinc blende (ZnS), Calamine (ZnCO_3), Zincite (ZnO). [Q521]
- Monazite = Thorium, Lanthanum, Cerium ka ore; beach sands mein milta hai. [Q519]
- Metallurgy = ores se metals ka extraction. [Q519]

2. Concentration & Dressing of Ores

- Concentration/Dressing = ore se impurities (gangue) remove karna; metal content increase karna. [Q522, Q524]
- Hydraulic washing/Gravity separation = heavy ore particles ko light gangue se alag karna water stream se; oxide ores ke liye. [Q524]
- Magnetic separation = magnetic ore ko non-magnetic gangue se magnet se alag karna; magnetite (Fe_3O_4) ke liye. [Q524]
- Froth flotation = sulphide ores ke liye; ore particles water mein oil/pine oil ke saath froth par aate hain. [Q524]
- Leaching = ore ko acid/alkali/bacteria se dissolve karke metal extract karna; bauxite (Al) aur gold ke liye. [Q524]

3. Roasting, Calcination & Smelting

- Roasting = ore ko air present mein strongly heat karna; sulphide ore $\rightarrow$ oxide; sulphur dioxide gas nikalta hai. Example: $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$. Purpose: sulphur remove karna. [Q520, Q523]
- Calcination = ore ko air absent mein strongly heat karna; carbonate ore $\rightarrow$ oxide; CO_2 aur water vapour nikalta hai. Example: $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$; $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O} \rightarrow \text{Al}_2\text{O}_3 + 2\text{H}_2\text{O}$. Purpose: CO_2 aur H_2O remove karna; volatile impurities remove. [Q520, Q525]
- Smelting = calcined/roasted ore ko reducing agent (coke/C, CO) ke saath heat karna furnace mein; molten metal aur slag alag hote hain. Example: $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$; $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$. [Q523, Q525]
- Roasting aur calcination mein ore ko solid state mein heat karte hain; smelting mein reduction hoti hai. [Q520, Q523]

4. Iron & Steel Production

- Iron mainly haematite (Fe_2O_3) se blast furnace mein extract hota hai. [Q526, Q528]
- Blast furnace charge: iron ore + coke + limestone. [Q526]
- Reactions: $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$; $\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$ (reducing agent). $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$. $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$; $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$ (slag). [Q528]
- Molten iron bottom par collect hota hai; slag upper layer par (lighter). [Q528]
- Pig iron = 3–4.5% carbon; hard, brittle; cast iron banta hai. [Q528]
- Wrought iron = purest form; <0.2% carbon; soft; wires, chains. [Q528]
- Steel = 0.2–2% carbon; alloying elements add karke stainless steel, tool steel, etc. [Q528, Q530]
- Bessemer converter ya Open-hearth furnace mein pig iron se steel banate hain. [Q530]
- Stainless steel = Fe + Cr + Ni + C; rust-resistant. [Q530]

5. Aluminium Extraction

- Aluminium ka main ore = Bauxite ($\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$). [Q519, Q527]
- Bayer's process se bauxite se Al_2O_3 (alumina) purify karte hain. [Q527]
- Hall-Héroult process: Molten alumina ka electrolysis cryolite (Na_3AlF_6) mein dissolved karke. Cathode: $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$. Anode: $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$; oxygen carbon anode se react karke CO_2 banata hai. Cryolite melting point kam karne aur conductivity badhane ke liye. [Q527]
- Aluminium good conductor, lightweight, corrosion-resistant; aircraft, utensils, electrical wires mein. [Q527]

6. Copper Extraction

- Copper ke ores: Copper pyrites (CuFeS_2), Copper glance (Cu_2S), Malachite. [Q521, Q529]
- Copper pyrites se extraction: Concentration by froth flotation. Roasting: $2\text{CuFeS}_2 + \text{O}_2 \rightarrow \text{Cu}_2\text{S} + 2\text{FeS} + \text{SO}_2$. Bessemerization: $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2$; phir $\text{Cu}_2\text{S} + 2\text{Cu}_2\text{O} \rightarrow 6\text{Cu} + \text{SO}_2$. Blister copper = 98–99% pure. [Q529]
- Electrolytic refining se 99.9% pure copper milta hai. [Q529]

7. ⚠ Traps / Key Differences Summary (Q519–Q531)

- ⚠ Trap: Bauxite = Aluminium ka ore; Hematite/Siderite/Magnetite = Iron ke ores; Monazite = Thorium ka ore — confuse mat hona. [Q519, Q521]
- ⚠ Trap: Roasting = air present mein, sulphide ore $\rightarrow$ oxide, sulphur remove hota hai. [Q520]
- ⚠ Trap: Calcination = air absent mein, carbonate ore $\rightarrow$ oxide, CO_2 aur H_2O remove hote hain. [Q520, Q525]
- ⚠ Trap: Roasting aur calcination mein ore solid state mein hota hai; dono heat treatment hain lekin atmosphere alag. [Q520, Q523]
- ⚠ Trap: Smelting = reduction furnace mein; reducing agent (coke/CO) use hota hai; metal + slag alag hote hain. [Q523, Q525]
- ⚠ Trap: Blast furnace charge = iron ore + coke + limestone; limestone flux ka kaam karta hai gangue ko slag mein convert karne ke liye. [Q526, Q528]
- ⚠ Trap: Pig iron (3–4.5% C) vs Wrought iron (<0.2% C) vs Steel (0.2–2% C) — carbon percentage se identify karte hain. [Q528]
- ⚠ Trap: Cryolite (Na_3AlF_6) electrolysis mein alumina dissolve karne ke liye use hota hai taaki melting point kam

ho; cryolite khud ore nahi balki added substance hai. [Q527]

- ⚠ Trap: Blister copper 98–99% pure hoti hai; "blisters" SO_2 gas bubbles ki wajah se bante hain. [Q529]